CRYSTALLOGRAPHY & X-RAY DIFFRACTION — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 16, 2026

Diagram-based questions on Crystallography & X-ray Diffraction

Question 1

10.0 points

Examine the diagram below illustrating the geometry of Bragg's law, where an incident X-ray beam strikes a crystal plane, and the diffracted beam is observed. Identify the angle of incidence, the spacing between planes, and the conditions for constructive interference according to the equation $n\lambda = 2d\sin\theta$. Explain how each component is represented in the diagram.

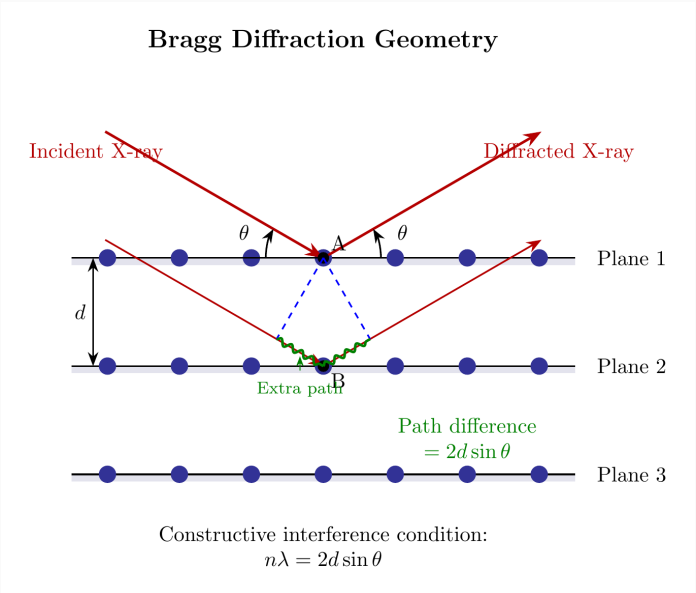

Fig. 1. MISMATCH [

Question 2 10.0 points

Examine the diagram below of the reciprocal lattice and Ewald sphere construction. Determine where the reflections will occur when an X-ray beam is incident on a crystal. Define the conditions under which a specific point in the reciprocal lattice satisfies the Laue condition, and interpret its meaning in the context of X-ray diffraction.

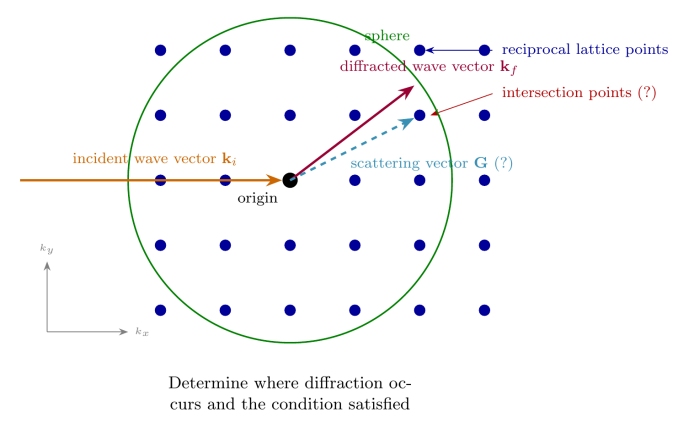

Fig. 2. A diagram showing the reciprocal lattice and Ewald sphere construction. Include the following elements: 1) A set of reciprocal lattice points arranged in a grid. 2) An Ewald sphere centered at the origin of the reciprocal lattice. 3) An incident X-ray beam represented as a vector pointing towards the Ewald sphere. 4) Highlight the points on the reciprocal lattice that intersect with the surface of the Ewald sphere, indicating potential reflection points. 5) Label the incident beam, Ewald sphere, and reciprocal lattice points. 6) Show the Laue condition as a vector from the origin to a lattice point on the sphere's surface. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Reflections • Ewald • Laue • forbidden reflections • equal areas The diagram will FAIL review if any of these appear. IMPORTANT: The diagram is missing required labels. Specifically: The diagram description requires showing and labeling "the Laue condition as a vector from the origin to a lattice point on the sphere's surface", but the. The diagram is missing a label for the vector representing the Laue condition, which is explicitly required by the diagram description.. Add each missing label as visible text at the appropriate location in the diagram. Ensure ALL labels are fully visible and not clipped by diagram boundaries. Use smaller font or reposition labels near edges if needed. IMPORTANT: The diagram is missing required labels. Specifically: missing label: Ewald sphere, missing label: reciprocal lattice points, missing label:. The diagram is missing required labels for the "Ewald sphere", "reciprocal lattice points", and the "Laue condition" vector, all of which are explicitly required by the diagram description.. Add each missing label as visible text at the appropriate location in the diagram. Ensure ALL labels are fully visible and not clipped by diagram boundaries. Use smaller font or reposition labels near edges if needed.

Question 3 10.0 points

Interpret the diagram below showing Miller indices (100), (110), and (111) planes in a cubic unit cell. Explain how these planes are oriented with respect to the cubic axes and how each set of indices determines the plane's orientation.

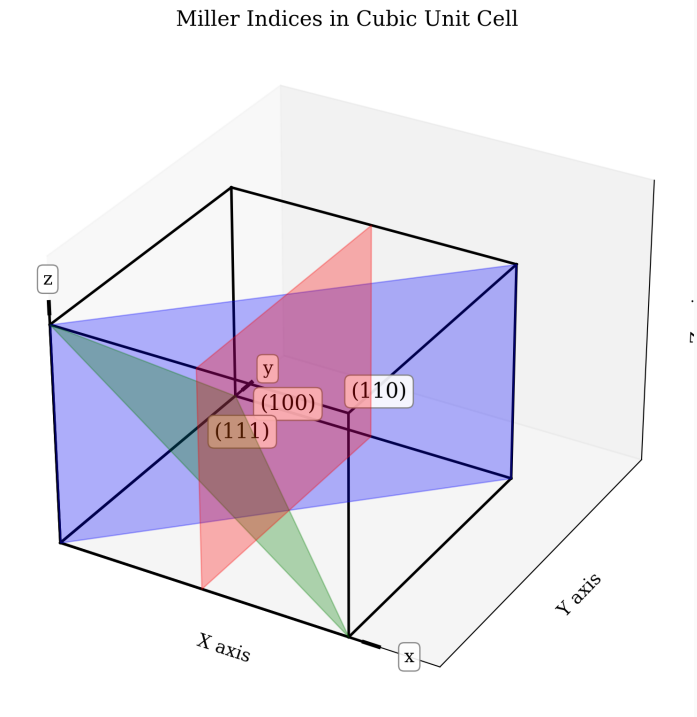

Fig. 3. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: Miller, equal areas; The diagram title "Miller Indices in Cubic Unit Cell" contains the forbidden label "Miller".. Remove the label 'Miller' from the title. The title should be "Indices in Cubic Unit Cell" or something

Question 4 10.0 points

Examine the FCC diffraction pattern in the diagram below. Identify the systematic absences in the diffraction peaks and explain the underlying cause of these absences based on the lattice plane reflections.

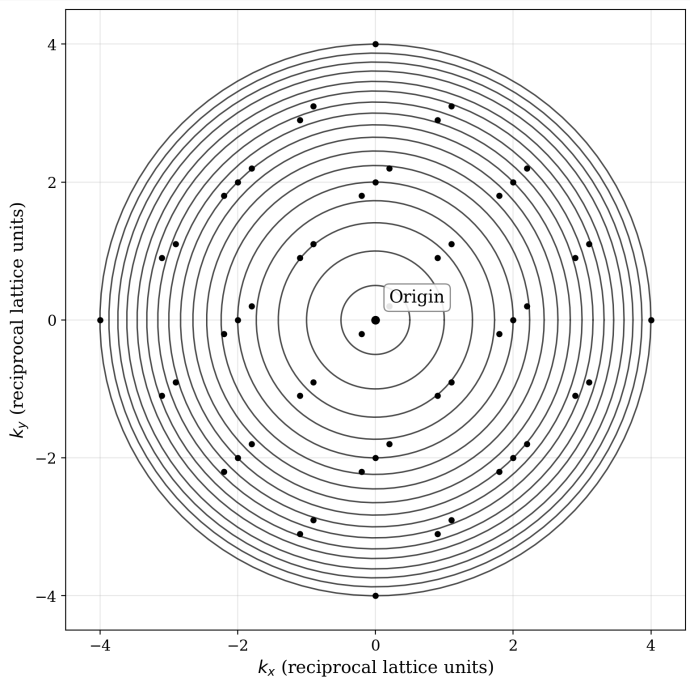

Fig. 4. CRITICAL: The previous diagram showed the wrong type. The diagram shows a series of concentric circles, but it completely omits the discrete diffraction peaks (dots) that are supposed to lie on these circles, which are essential for representing a diffraction pattern and answering the question.. Specifically: The diagram is missing the required diffraction peaks (small dots. Regenerate from scratch showing the CORRECT diagram type. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: FCC, Systematic, base editing window, forbidden reflections, systematic absences; The diagram includes the forbidden label "FCC" in its title, which directly violates the review instructions.. A 2D plot representing a diffraction pattern. The horizontal axis should be labeled 'kx (reciprocal lattice units)' and the vertical axis 'ky (reciprocal lattice units)'. The plot should show a series of concentric circles centered at the origin. Small dots, representing diffraction peaks, should be placed on some of these circles, consistent with the pattern for a face-centered cubic structure. The diagram should visually imply the presence of systematic absences by having some circles without any diffraction peaks on them. A central point should be labeled 'Origin'. Do not add a title to the diagram.

Question 5 10.0 points

Examine the atomic arrangement of the BCC crystal in the diagram below. Use this to calculate the structure factor for a given set of Miller indices, assuming only atomic scattering factors are considered. Detail the steps involved in the calculation.

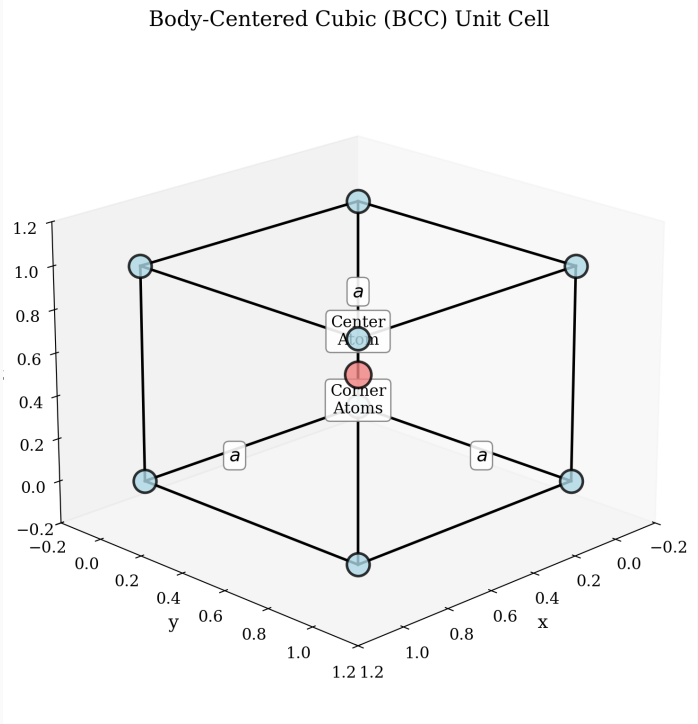

Fig. 5. 3D BCC unit cell showing atoms at lattice positions. The unit cell is a cube with atoms at each corner and a single atom at the center. Highlight the cubic structure and atomic positions to aid in understanding the BCC arrangement for structure factor calculation. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Step • Identify • Apply • Result • Dependent • Miller The diagram will FAIL review if any of these appear.